SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Faculty: Dr. V. Parthipan

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **S. Bharath kumar (192511126)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Bharath kumar

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _______→ Transport → Network → _______→ Physical.

analyse the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyse how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl→DataLinkLayer→ErrorDetection→ Transport Layer → Error Correction

analyse how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyse how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

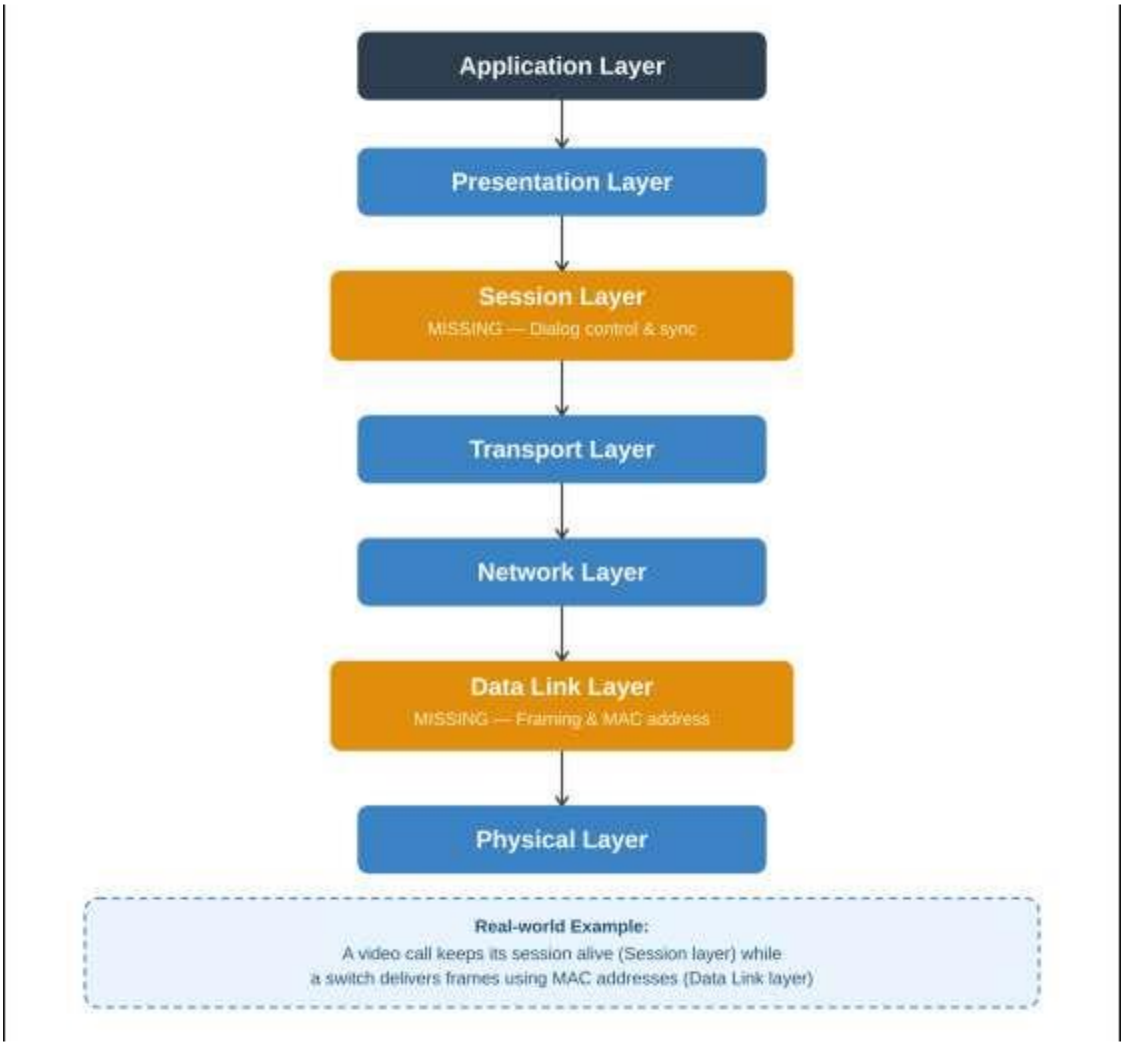
Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

analyse the problem and explain how the concept map helps in troubleshooting.

Annexure A

Concept Mapping: Answers

Q1. Missing Layers in the Concept Map



Application → Presentation → Session → Transport → Network → Data Link → Physical.

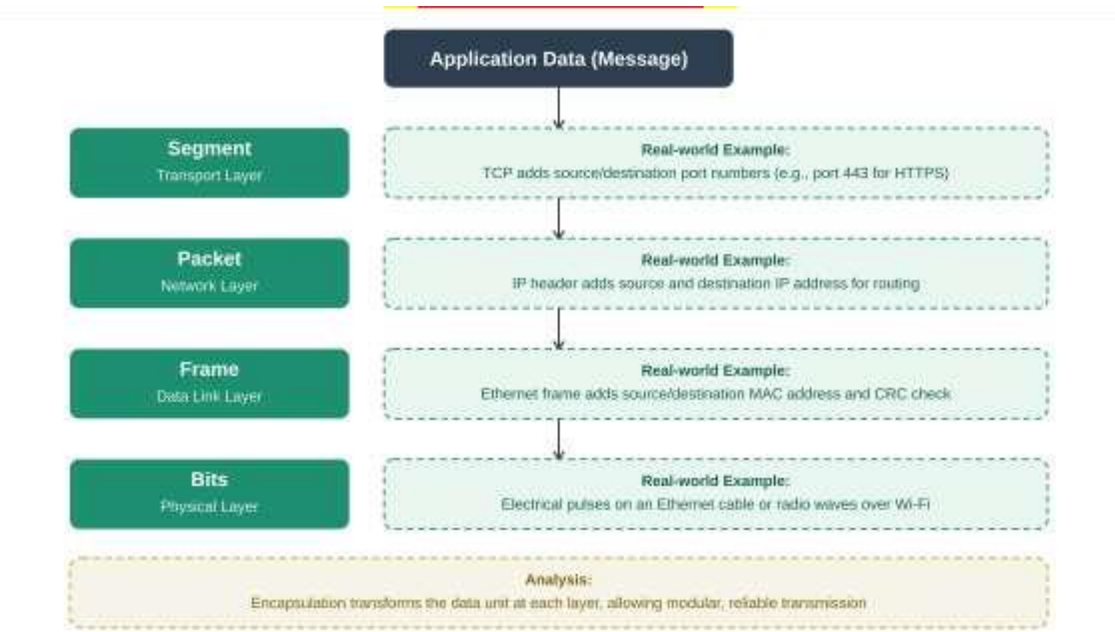
The two missing layers are the Session Layer and the Data Link Layer.

- **Session Layer**: Establishes, manages, and terminates the communication sessions between two communicating applications. It handles dialog control (deciding whose turn it is to transmit), synchronization (inserting checkpoints so that, in case of a crash, only data after the last checkpoint needs retransmission), and session recovery. Without this layer, applications on either end would have no organized way to open, maintain, and close a logical connection, making multi-request exchanges (like a long file transfer or a login session) unreliable.

- **Data Link Layer**: Provides node-to-node delivery of data within the same network segment. It is responsible for framing (organizing bits into frames), physical addressing (MAC addresses), error detection/correction at the frame level (using CRC), and flow control between adjacent nodes. It converts the raw bit stream from the Physical layer into a structured frame that the Network layer can use, and it detects transmission errors introduced by the physical medium.

Together, these two layers are essential for reliable communication: the Session layer ensures orderly, synchronized, and recoverable dialogues at the top of the stack, while the Data Link layer ensures that data survives the noisy physical medium and reaches the correct adjacent device error-free. Their absence would break both the logical organization of communication and the physical reliability of frame delivery.

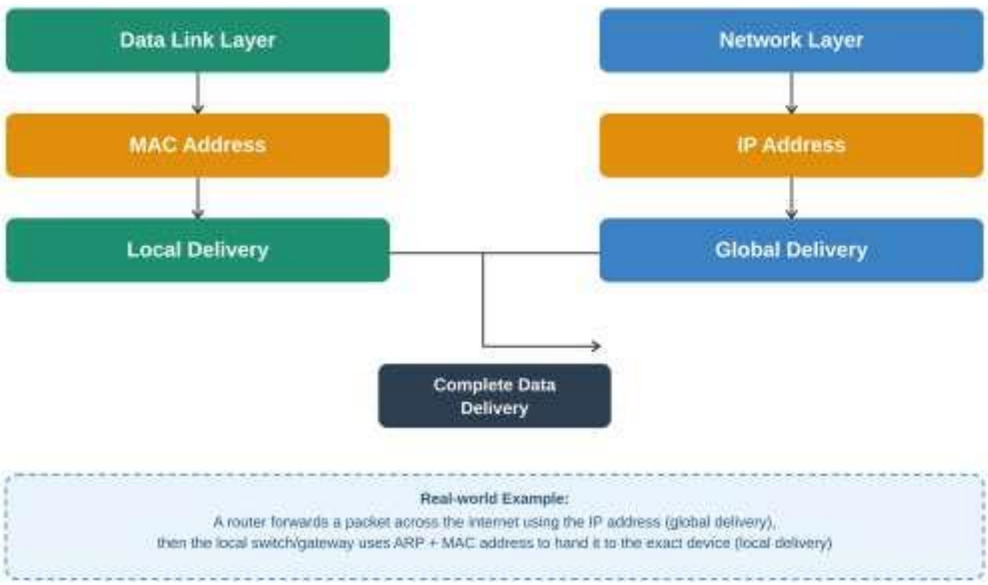
Q2. Concept Map: Data Units and OSI Layers



Analysis: As data moves down the OSI stack, each layer adds its own control information (a process called encapsulation) and transforms the data unit into a new form. This transformation is what supports transmission: a Segment cannot travel across a network without a network address, so it must become a Packet; a Packet cannot be delivered to a specific device on a

LAN without a MAC address and framing, so it must become a Frame; and finally, no device can physically send a Frame without converting it into Bits for the transmission medium. At the receiving end, the reverse process (decapsulation) strips each header away layer by layer. This layered transformation ensures that each layer only needs to understand its own data unit and header, which keeps the design modular, scalable, and reliable.

Q3. MAC Address (Local Delivery) and IP Address (Global Delivery)

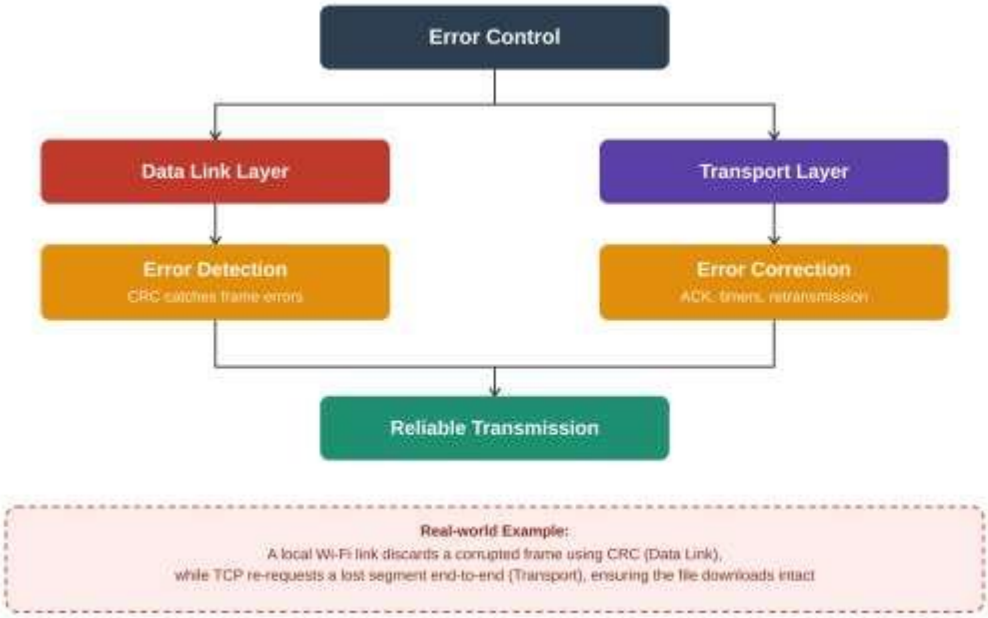


The MAC address is a physical, hardware-burned address used by the Data Link layer to deliver frames between devices on the same local network segment (LAN). It is only meaningful within that local broadcast domain and is used, for example, by switches to forward frames to the correct port.

The IP address is a logical address used by the Network layer to deliver packets across multiple interconnected networks, including the wider internet. Routers use the destination IP address to determine the best path across networks, moving the packet from one network to the next until it reaches the destination network.

These two relationships work together to ensure complete, end-to-end data delivery: the IP address gets the packet to the correct destination network (global delivery), and once the packet arrives at that network, the MAC address is used (via ARP, Address Resolution Protocol) to deliver it to the exact device on that local segment (local delivery). Without the IP address, a router could not decide which network to forward data to; without the MAC address, a switch could not deliver the frame to the correct device once it arrives locally. Both addressing schemes are therefore complementary and both are required for successful communication from source to destination.

Q4. Error Control: Data Link Layer (Detection) and Transport Layer (Correction)

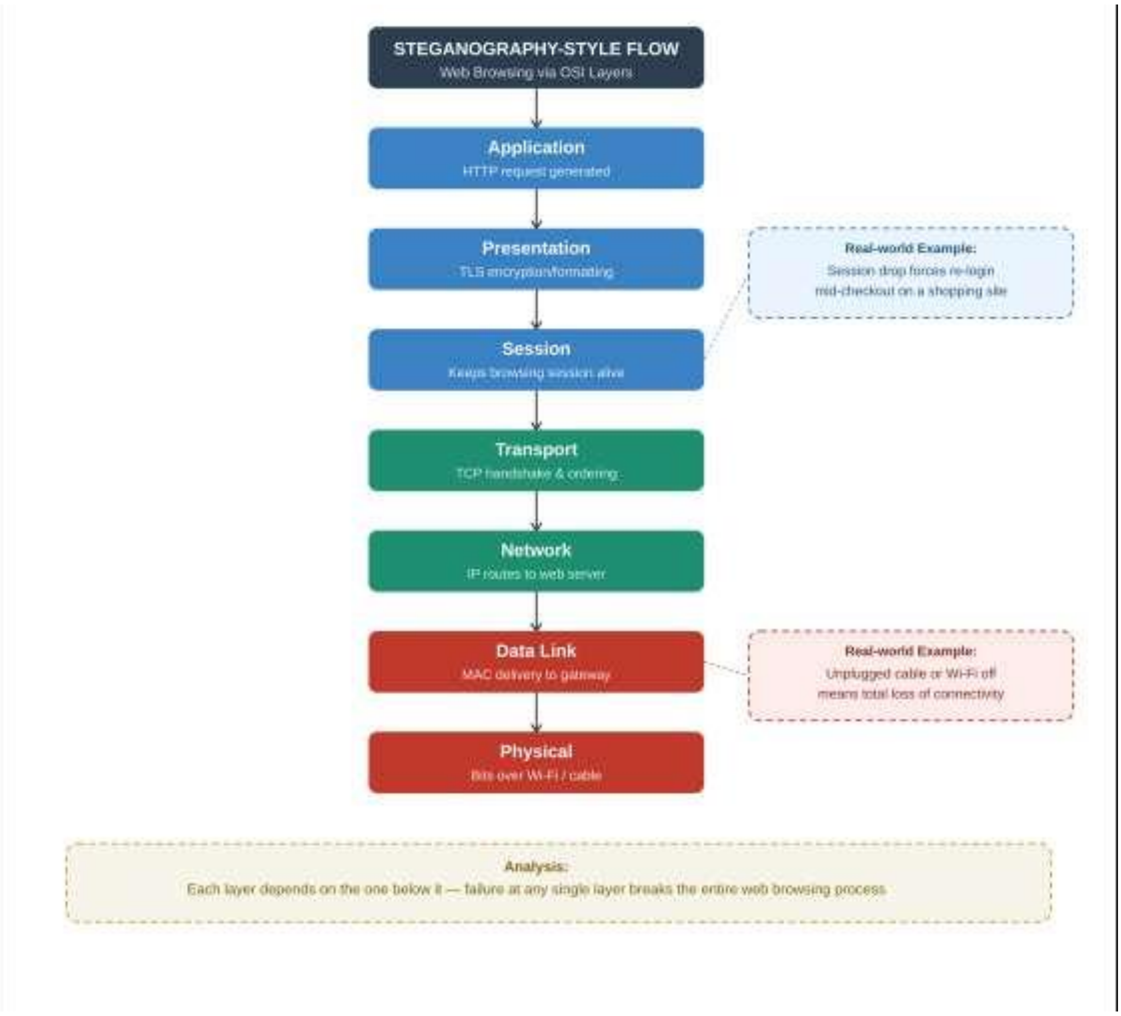


The Data Link layer performs error detection at the frame level, typically using techniques such as Cyclic Redundancy Check (CRC) or checksums. When a frame arrives, the receiving device recalculates the check value and compares it with the one sent; if they differ, the frame is known to be corrupted. However, the Data Link layer on a shared or point-to-point local link usually only detects the error (or in some protocols, requests retransmission from the immediate neighboring node) rather than performing full end-to-end correction.

The Transport layer (e.g., TCP) is responsible for reliability across the entire end-to-end path, which may include multiple hops and networks. It performs error correction and recovery in a broader sense through mechanisms such as sequence numbers, acknowledgments (ACKs), timers, and retransmission of lost or corrupted segments. This ensures that even if an error slips past a lower layer, or a packet is lost or arrives out of order somewhere along the path, the Transport layer will detect the gap and request the correct data again.

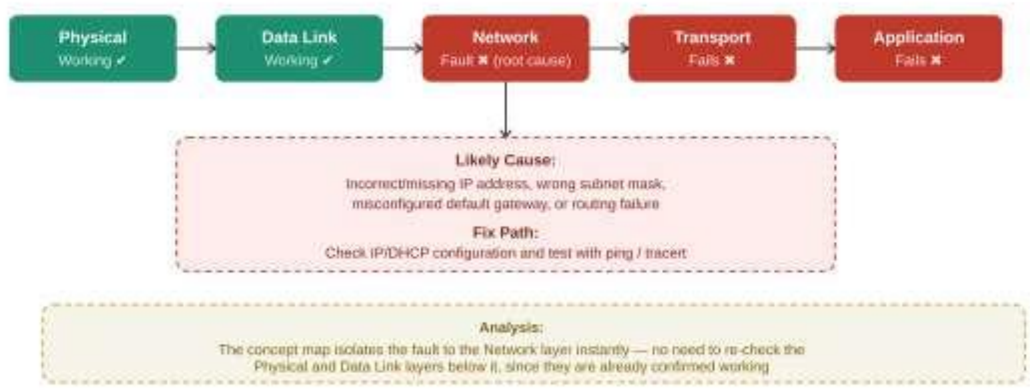
Analysis: The interaction between these two layers creates a layered defense for reliability. The Data Link layer catches and discards corrupted frames early, close to where the error occurred, preventing bad data from wasting bandwidth further down the path. The Transport layer then provides a safety net over the entire connection, guaranteeing that the application ultimately receives complete and correctly ordered data even if the Data Link layer at some intermediate hop failed to catch an error or a frame was dropped. This layered approach (local detection + end-to-end correction) is more efficient and more reliable than depending on a single layer to handle all error control.

Q5. Concept Map: Web Browsing Using OSI Layers



Analysis: Web browsing depends on every layer functioning correctly in sequence, because each layer builds on the service provided by the one below it. A failure at any single layer breaks the entire chain of communication, even if all other layers are working perfectly — for example, a perfectly good HTTP request (Application layer) is useless if the Physical layer has no signal, and a strong physical connection is useless if the Network layer cannot route the packet to the correct server. This illustrates why OSI is described as a layered, dependent model: reliable web browsing is only possible when Application-layer requests, Presentation-layer formatting, Session-layer continuity, Transport-layer reliability, Network-layer routing, Data Link-layer local delivery, and Physical-layer signal transmission all succeed together.

Q6. Troubleshooting Using the Concept Map



Analysis of the problem: Since the Physical and Data Link layers are working (✓), this means the device has a valid electrical/wireless signal and can successfully communicate with its immediate neighboring device (e.g., the local switch or Wi-Fi access point) using MAC addressing. However, the failure begins at the Network layer (✕), and everything above it (Transport ✕, Application ✕) also fails as a direct consequence.

This pattern strongly indicates a problem at the Network layer itself, most likely one of the following: an incorrect or missing IP address/subnet configuration, a misconfigured default gateway, a routing failure, or a DNS/IP addressing conflict. Because the Network layer cannot route or address the packet correctly, no packet can ever reach the Transport layer for connection establishment, and therefore no Application-layer request (like a webpage or email) can succeed even though the physical connection itself is completely healthy.

How the concept map helps in troubleshooting: The concept map allows a network engineer to isolate the fault quickly by checking layers in a bottom-up sequence, exactly like the OSI troubleshooting model. Since the two lowest layers are confirmed working, effort should not be wasted checking cables, Wi-Fi hardware, or switch ports. Instead, the technician should immediately focus on Network-layer issues: verifying IP address assignment (e.g., via DHCP), checking the subnet mask and default gateway, and testing basic reachability (e.g., using ping or tracert). This layer-by-layer visualization prevents wasted effort on layers that are already functioning and directs attention precisely to the point of failure, making diagnosis faster and more systematic.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation